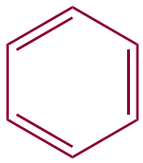
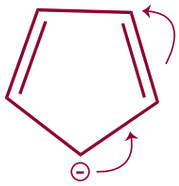
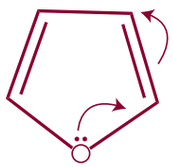
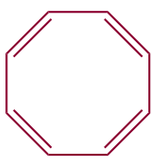
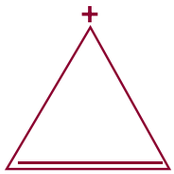
Some of the examples for Huckel rule

1.	 **Benzene**	(i) The benzene is a planar molecule (ii) It has six deloclaised π electorns (iii) 4n + 2 = 6 4n = 6- 2 n = 1 it obeys Huckel's (4n+2) π electron rule with n = 1.hence, benzene is aromatic.
2.	 **Cyclo penta dienyl anion**	(i)cyclopenta dienyl anion a planar ring structure (ii) It has 6 delocalised electrons (iii) 4n + 2 = 6 n = 1 it obeys Huckel's (4n+2) π electron rule with n = 1.hence, it is aromatic.
3.	 **Furan**	(i) Furan has a planar ring structure. (ii) if has 6 delocalised electrons (iii) 4n+2 = 6 n=1 it obeys Huckel's (4n+2) π electron rule with n = 1.hence, it is aromatic.
4.	 **Cyclo penta diene**	(i) It has planar structure (ii) It has four π electron but the π electrons are not delocalised and hence it is not an aromatic compound
5.	 **Cyclooctatetraene**	molecule is non planar and hence it is not an aromatic compound **Cyclooctatetraene**
6	 **Cyclopropenylcation**	(i) cyclopropenyl cation has planar structure (ii) It has 2 delocalised π electron. (ii) 4n + 2 = 2 4n = 0 n = 0 (an interger) and hence it is aromatic compound.

Unit 13 - jagan.indd 206 22-01-2021 12:51:38